Test report 22 – studio look (lights, soft shadows, ambient occlusion, ground)

2026-10-27 · Branch claude/cool-edison-1goqgj · Commit 8989456 · mcad 0.3.0 · rustc 1.97.0 (2d8144b78 2026-07-07) · 4 cores. CLI measurements with the release binary without daemon (`tools/report/report22-data.py`, median of 5 runs), UI with Playwright (`web/e2e/studio.spec.ts`, German UI as in all specs), test runs on the same commit (`tools/report/report22-tests.sh`).

Mode	Settings	Render time (1024×768)	cargo test
--style studio	**11 + background**	**3.6× standard**	**695 / 695**
CLI, UI toggle “Studio look”, JSON `style/studio`	light, shadows, AO, ground, background – same defaults in CLI and UI	median over 6 examples; +898…1,138 ms	clippy clean, fmt clean

Vitest · Playwright
271 · –
studio spec 0 / 0

1 What was built

Commit	Title
`682d280`	render: studio style with lights, soft shadows, ambient occlusion and ground
`a02ce6e`	ui: studio look in the 3D view (lights, soft shadows, ambient occlusion, ground)
`8989456`	report 22: generators for the studio look report; e2e S06 English UI

CLI: software renderer (`crates/cad-render/src/studio.rs`)

- **Deferred shading.** With `--style studio` the rasteriser writes albedo and the interpolated (camera-facing) normal per supersample into a G-buffer instead of a shaded colour; section caps go in with their hatched colour. After the caps one parallel pass (rows over all cores, `std::thread::scope`) lights every opaque sample; edges, transparent bodies, overlays, labels and the axis triad are drawn afterwards as before.
- **Lights.** Key light (Lambert + Blinn-Phong highlight, exponent 48) whose azimuth is relative to the camera like a turntable and whose elevation is measured from the xy plane, a weak fill light from the camera (+15°), hemisphere ambient light (white sky from +z, bounce tinted by the ground colour). A soft shoulder above 0.8 keeps directly lit light faces from burning out.
- **Shadows.** Orthographic shadow map (2048², span-based rasteriser) of all opaque, clipped triangles along the key light; percentage-closer soft shadows: blocker search (12 Vogel samples) → penumbra from the blocker–receiver distance × tan(light radius) → 16 filter samples, rotated per sample by interleaved gradient noise, compared against the receiver plane (depth gradient from the normal) instead of one depth, plus normal offset. Tiles of 16×16 texels with their minimum depth skip the search where nothing can cast. Model on itself and on the ground; caps and transparent bodies cast none.
- **Ambient occlusion.** Screen space from the linear depth buffer (including the ground): 16 hemisphere samples of radius `--ao-radius` (default 5 % of the model diagonal) around the normal, rotated in a 4×4 pattern, range check, then a depth- and normal-aware 4×4 blur. AO darkens the ambient light fully and the direct light by 55 %.
- **Ground and background.** Plane z = lowest point of the model, drawn only where the camera looks down onto it (none in front/right/bottom views), not in the depth buffer of the overlays (grids and labels stay visible), faded into the background between 0.55 and 1.5 model diagonals. Background solid or vertical gradient (`--background`, also usable with `standard`). Edges at 35 % opacity.

- **Interfaces.** `RenderOptions::studio: Option<Studio>`, `background_bottom`, `CameraInfo::studio` (light vector, ground height, AO radius, shadow map size); CLI options in `ViewArgs` (also `mcad watch --render`; the daemon forwards the arguments), JSON `style`, `studio`, `background`, `views[].studio`. Studio options without `--style studio`, out-of-range values and unknown colours exit with 2.

UI: three.js (`web/src/view/viewer.ts`, `web/src/app/studio.tsx`, `web/src/logic/studio.ts`)

- Toggle “Studio look” and a settings panel in the view toolbar; settings in the store, kept in localStorage `cad-studio` (sanitised on load), same defaults as the CLI (Vitest checks both and the generated CLI arguments).
- Directional key light with a 2048² variance shadow map (blur radius from the softness, as the CLI's penumbra at a typical distance), fill light and hemisphere light placed every frame from the camera azimuth (`lightDirection`, same convention as the Rust code); opaque meshes cast and receive shadows; materials a little glossier (roughness 0.5) and edges at 35 % opacity.
- Ground: a transparent `MeshStandardMaterial` plane with a radial fade (shader chunk), receiving shadows; the floor grid is hidden while it is on. Background: a procedural 1×256 canvas gradient as `scene.background` (no image files).
- Ambient occlusion: three.js `GTAOPass` (no new dependency) on the opaque model and the ground only; frame = model pass to the screen, AO multiplied on top, then the overlays (handles, grids, measurements, sketches, asserts) drawn over the kept depth buffer, so they are not darkened. Paused while a section is shown.
- “Copy as mcad render” gives the CLI call of the current view (perspective, azimuth, elevation, theme) with the changed settings; `mcad ui status` reports `studio` with that call.

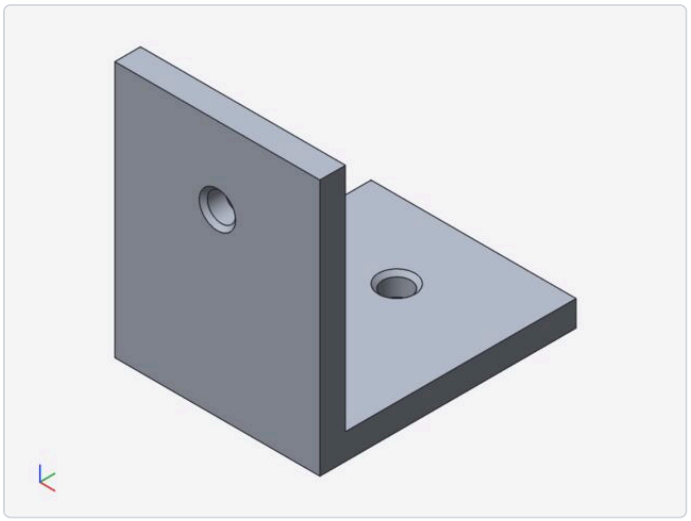
2 Settings

Setting	CLI	UI (English)	Default	Range, notes
Style	`--style standard|studio`	toggle “Studio look” (sparkles icon)	standard	–
Key light direction	`--light AZ,EL`	Direction, Height	−40°, 50°	azimuth relative to the camera (0 = from the camera, + = from the right); elevation 1…90° above the xy plane
Light intensity	`--light-intensity F`	Intensity	1	0…4 (UI 0…2.5)
Ambient (sky) share	`--ambient F`	Ambient	0.4	0…1; the rest is key light
Shadows	`--shadows on|off`	Shadows	on	model on itself and on the ground
Shadow softness	`--shadow-softness F`	Softness	0.35	0 hard … 1 very soft (light radius 0…5°)
Ambient occlusion	`--ao on|off`	Ambient occlusion	on	corners, gaps, contact with the ground
AO radius	`--ao-radius MM`	Radius (empty = auto)	5 % of the model diagonal	> 0 mm
AO strength	`--ao-strength F`	Strength	1	0…2
Ground	`--ground on|off`	Ground	on	plane at the lowest z of the model, fades out; only seen from above
Ground colour	`--ground-color C`	colour field, “Theme”	light `#e6e8ec`, dark `#2c3038`	`#rrggbb`, `rrggbb`, palette name
Background	`--background C[,C]`	two colour fields, “Theme”	studio light `#f8f9fb`→`#d9dde3`, dark `#2b3038`→`#111317`	one colour or top,bottom gradient (also for `standard`)

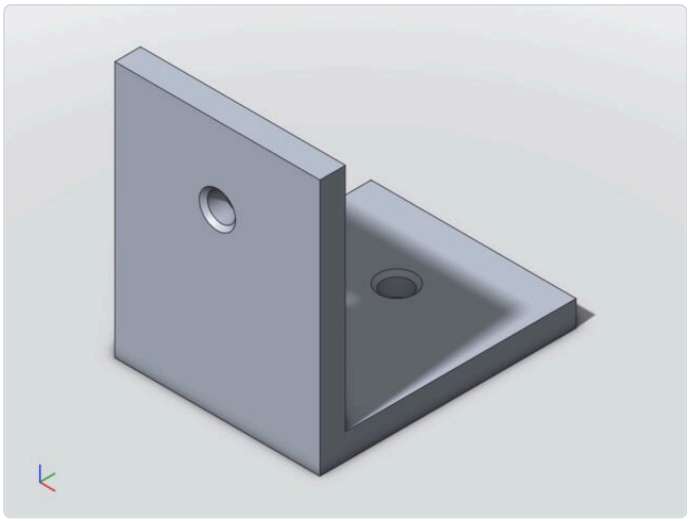
3 CLI screenshots

All images `--size 900x675` (contact sheet 1600×1200), iso view unless stated; the command of each image is in `data.json`, e.g. `mcad render examples/05-box-lid.mcad -o 05-box-lid-studio.png --size 900x675 --style studio`.

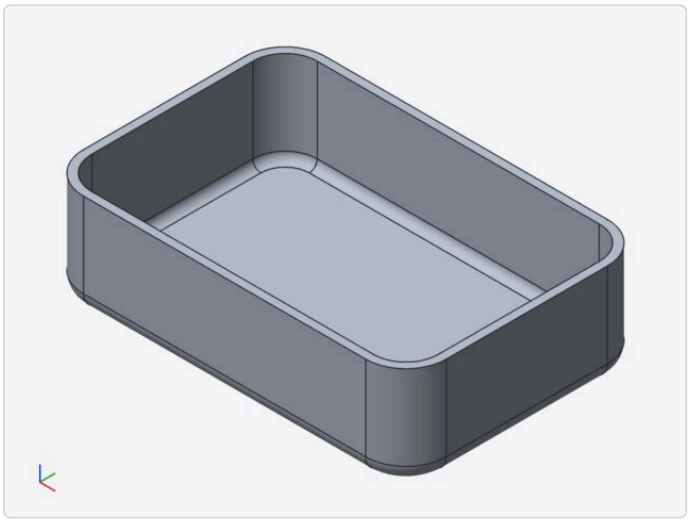
Standard and studio



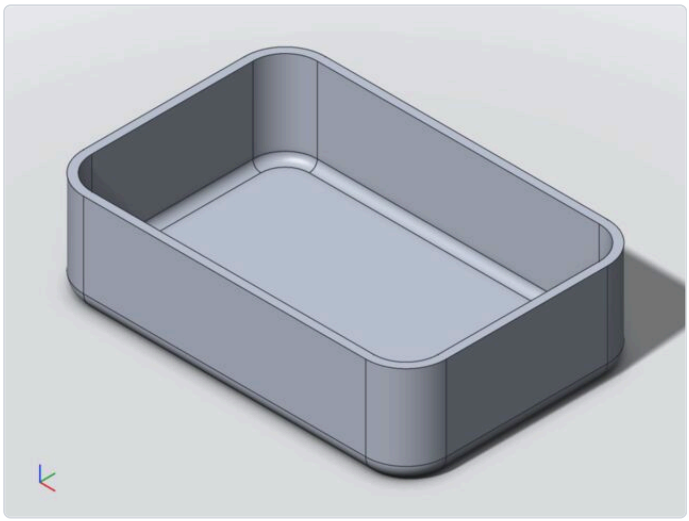
`02-bracket` standard



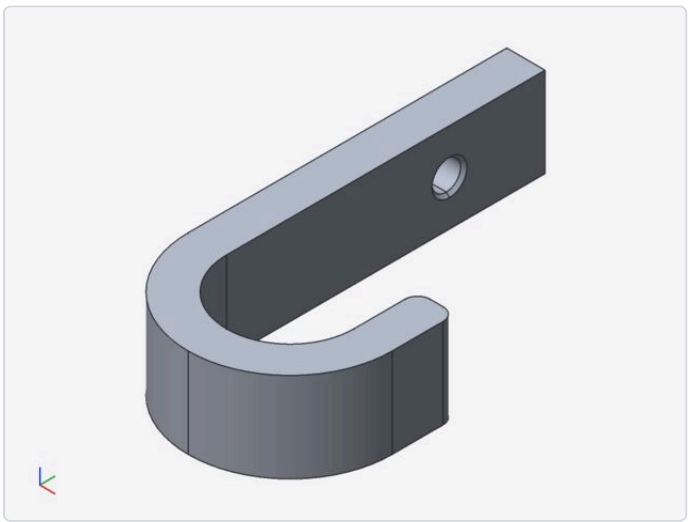
`02-bracket` studio (88 % of the pixels differ)



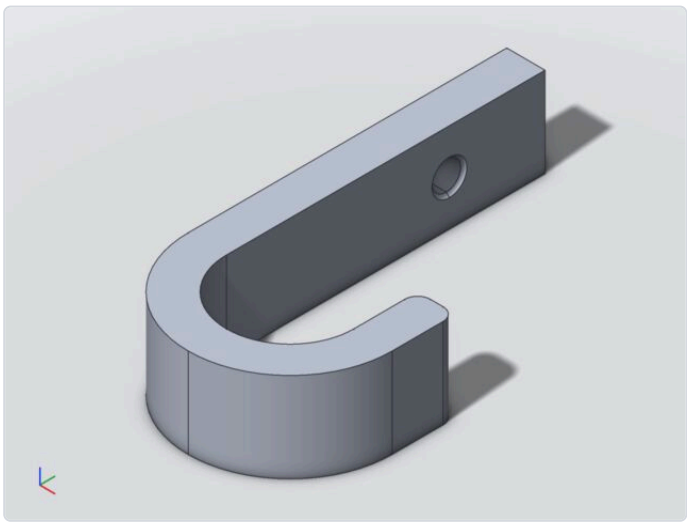
`03-tray` standard



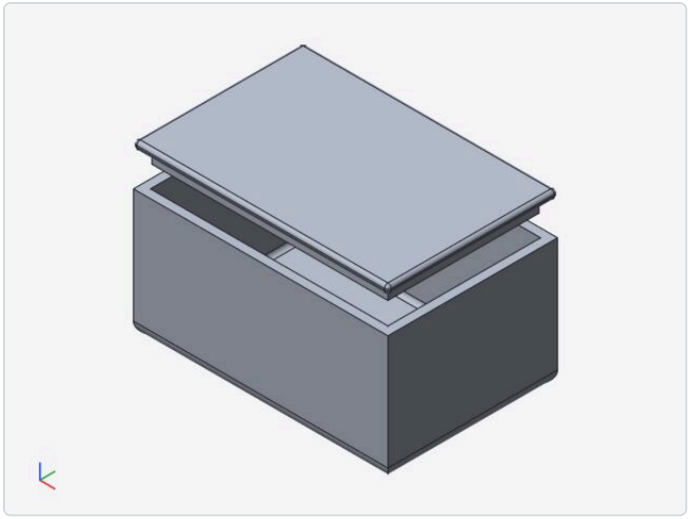
`03-tray` studio (77 % of the pixels differ)



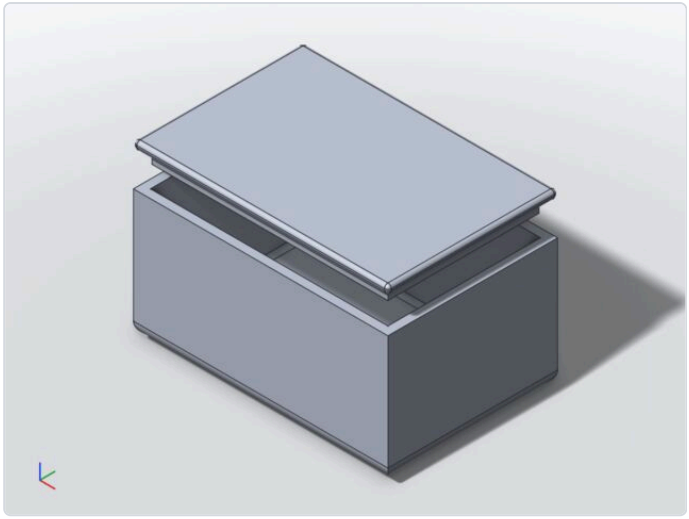
`04-hook` standard



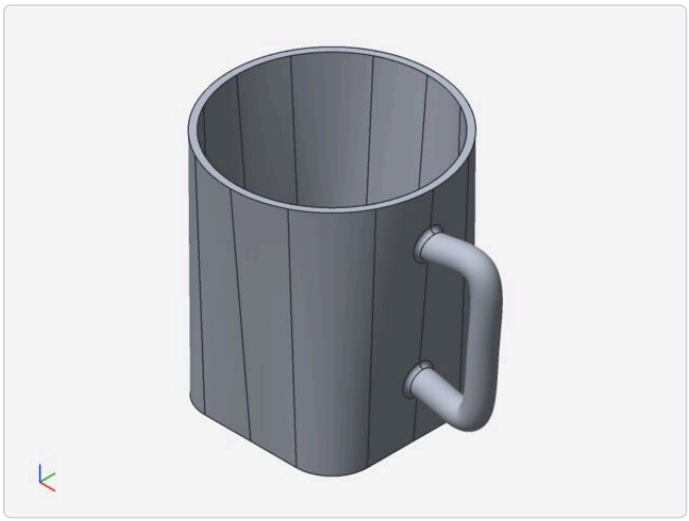
`04-hook` studio (81 % of the pixels differ)



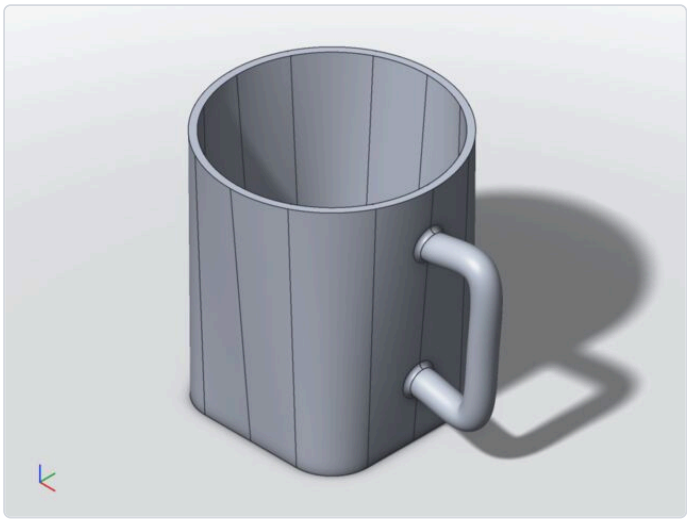
05-box-lid standard



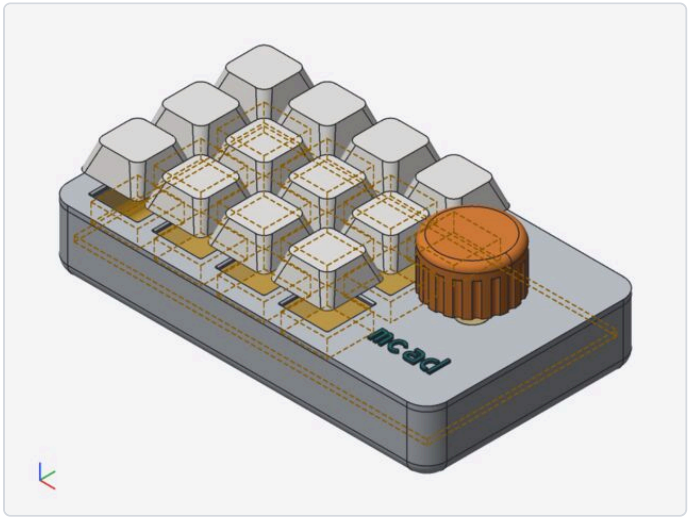
05-box-lid studio (61 % of the pixels differ)



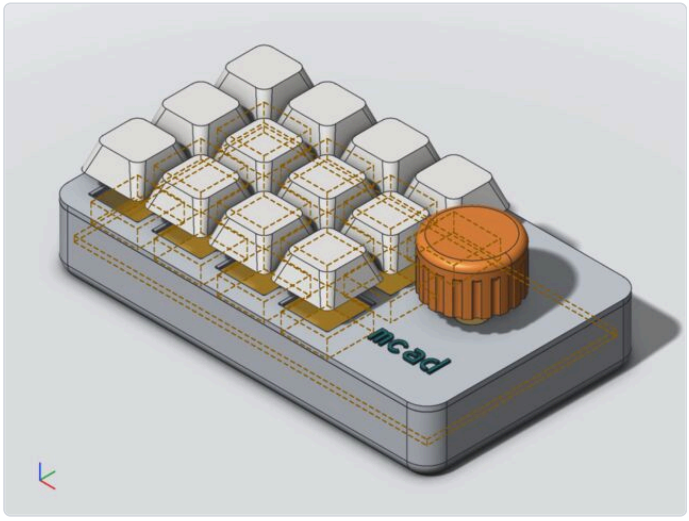
07-mug standard



07-mug studio (78 % of the pixels differ)

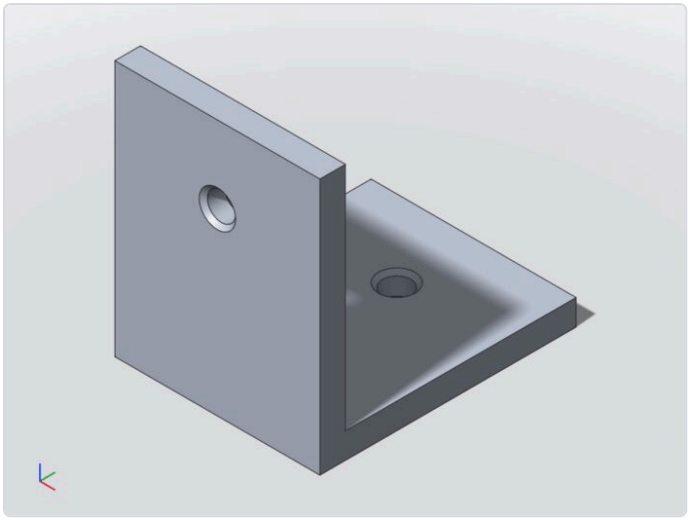


08-macropad standard

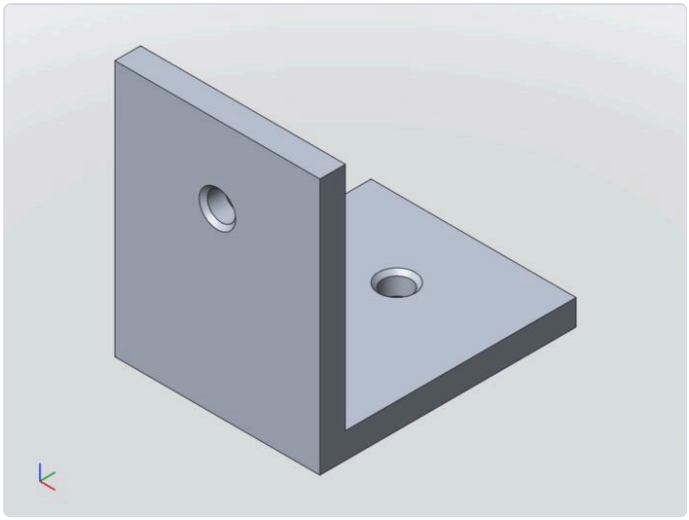


08-macropad studio (77 % of the pixels differ)

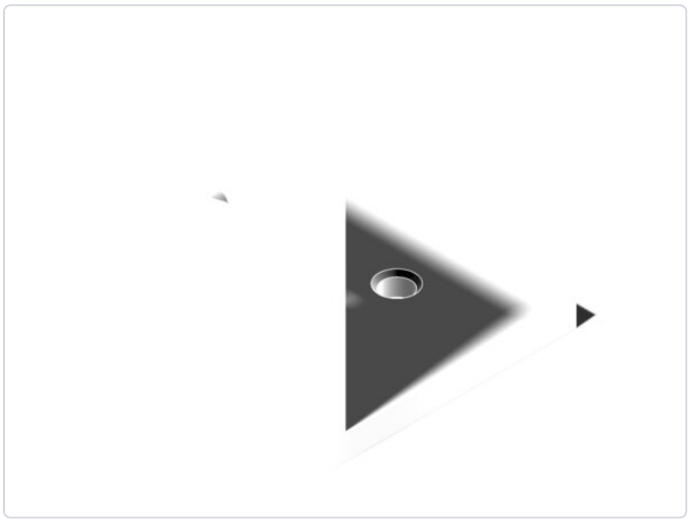
Shadows on and off



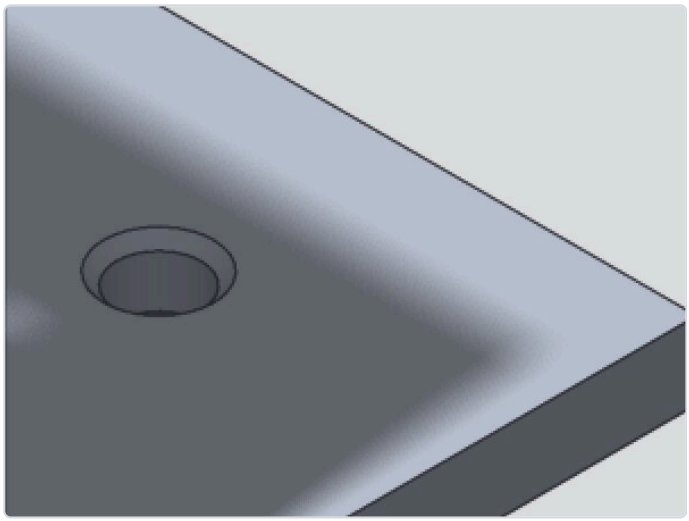
Shadows on (AO off): the upright throws a soft shadow onto the base, light shines through its bore; mean brightness of the base probe 133



`--shadows off`: probe 191; 6.3 % of the pixels differ

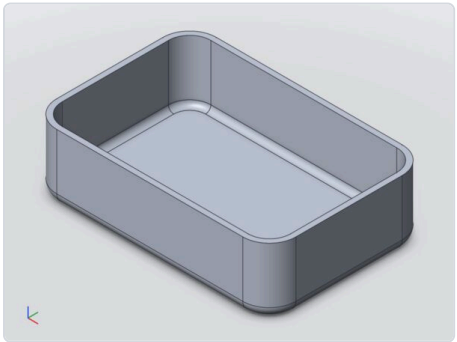


Difference shadows on − off, ×2, inverted (dark = changed): the shadow of the upright on the base and on the ground

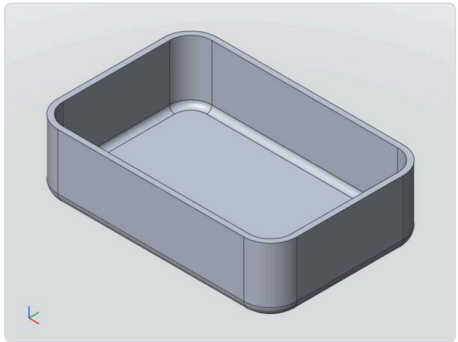


Shadows on, crop ×2: penumbra widens away from the contact

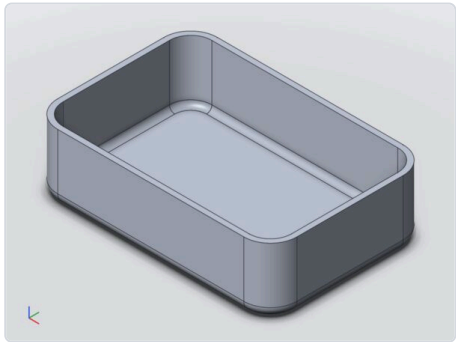
Ambient occlusion on and off



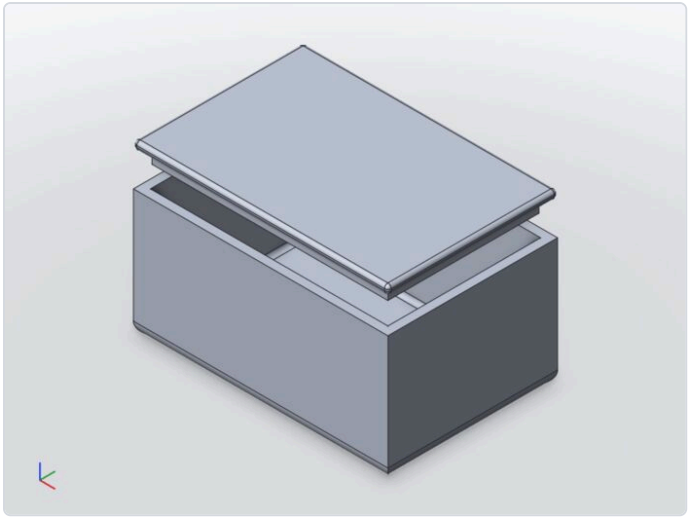
Tray, AO on (shadows off)



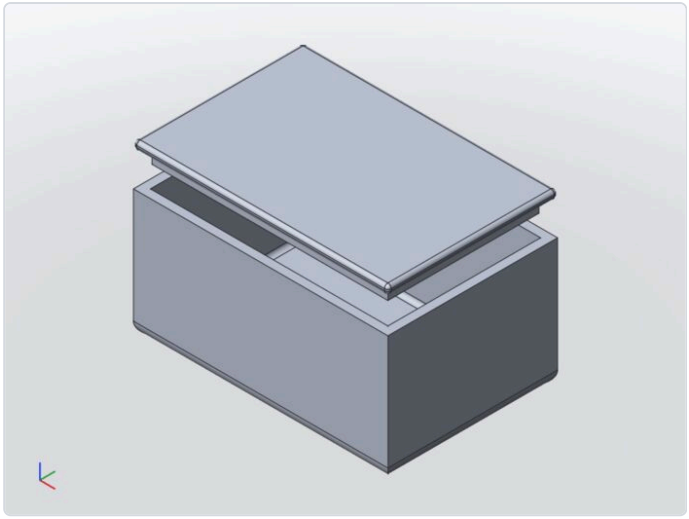
AO off: 2.8 % of the pixels differ



`--ao-strength 2 --ao-radius 6`: 6.1 % differ from AO off

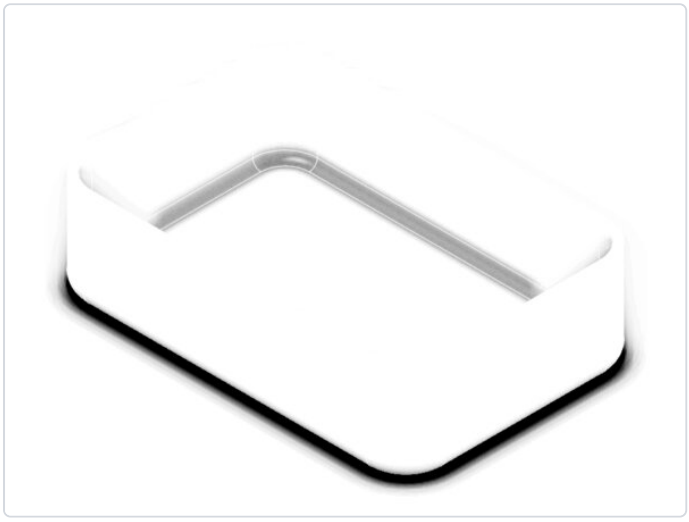


Box, AO on: inner corners, the gap under the lid and the contact with the ground darken

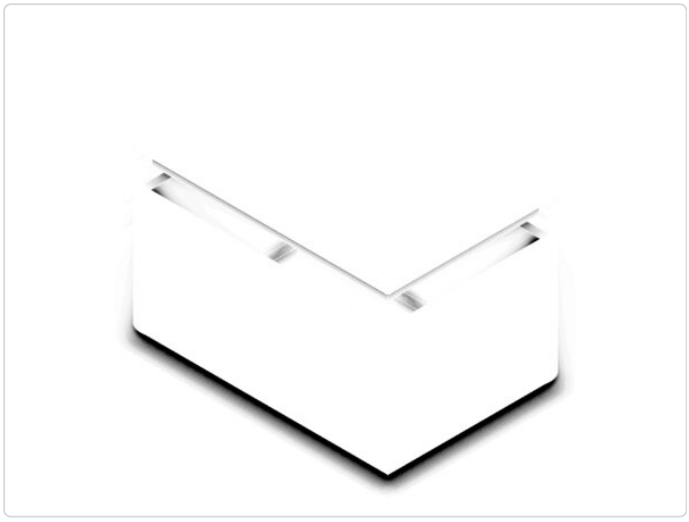


AO off (1.8 % differ)

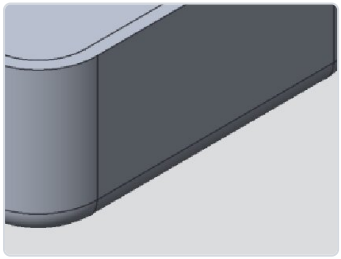
Ambient occlusion is deliberately subtle at the default strength; the amplified differences show where it acts, the crops (×2, nearest neighbour) show it at the ground contact and in the inner corner.



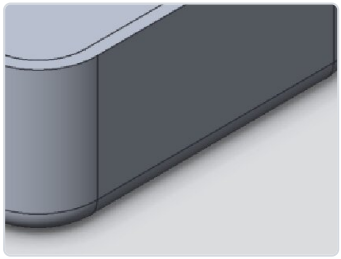
Tray: difference AO on − off, ×8, inverted: inner wall corners, the floor fillet, the ground contact



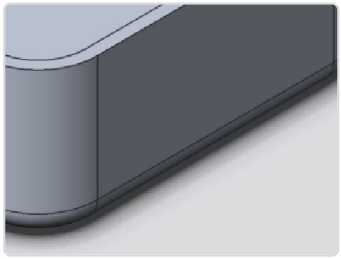
Box: difference AO on − off, ×8, inverted: the gap under the lid, inner corners of the bin, the ground contact



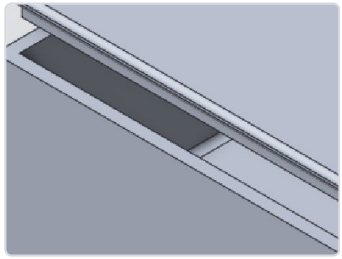
Tray, AO off (crop at the ground contact)



AO on (default)

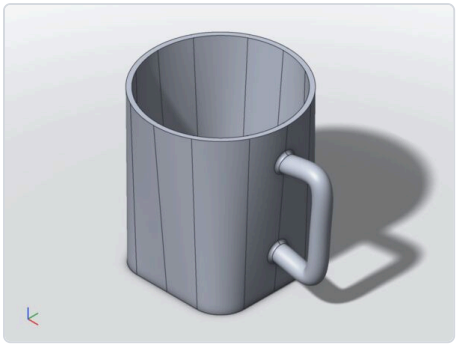


`--ao-strength 2 --ao-radius 6`



Box, AO on: gap under the lid

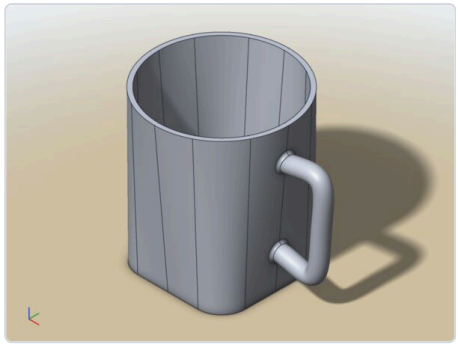
Ground and background



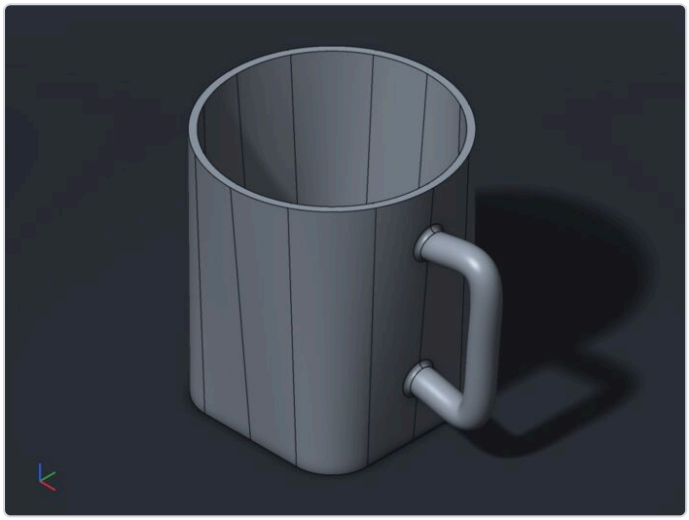
Default: ground fades into the light gradient



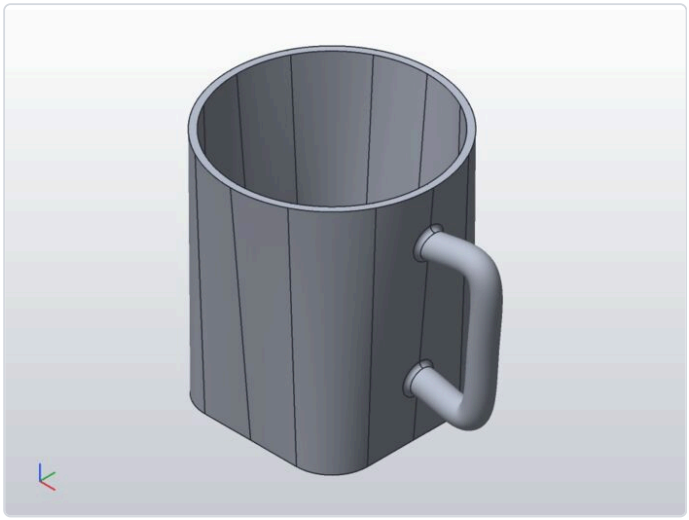
`--ground off --background ffffff`



`--ground-color d8c8a8 --background eef3f8,9fb3c8`

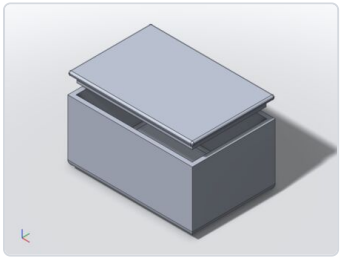


`--theme dark`: ground and gradient from the dark theme

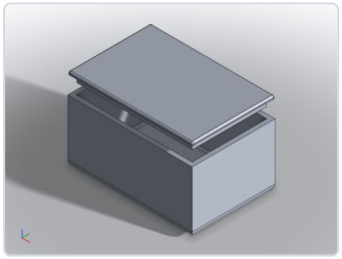


`--background ffffff,c8ccd2` with the standard style

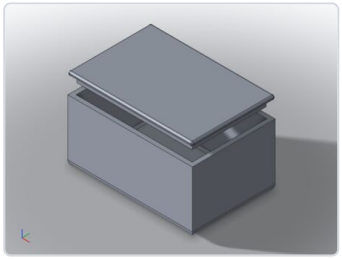
Light direction, intensity, softness



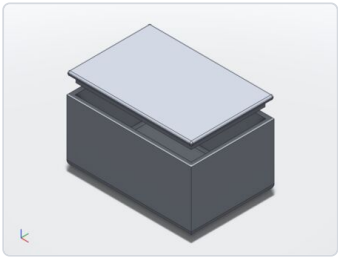
`--light -40,50` (default)



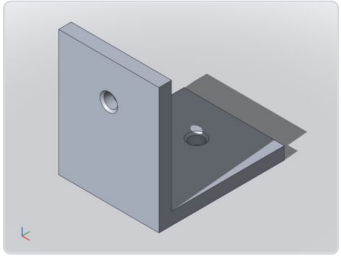
`--light 60,25`



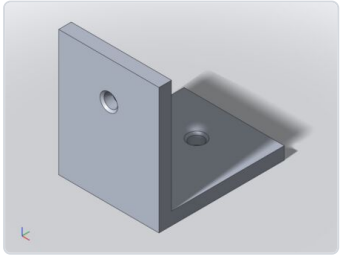
`--light -100,20`



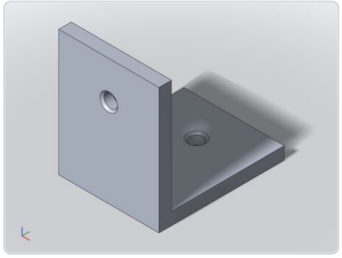
`--light 0,85`



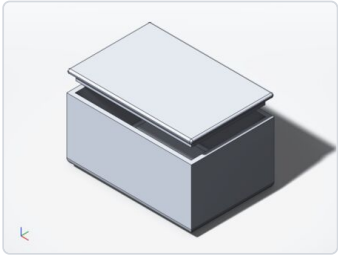
`--shadow-softness 0`



`0.35` (default)

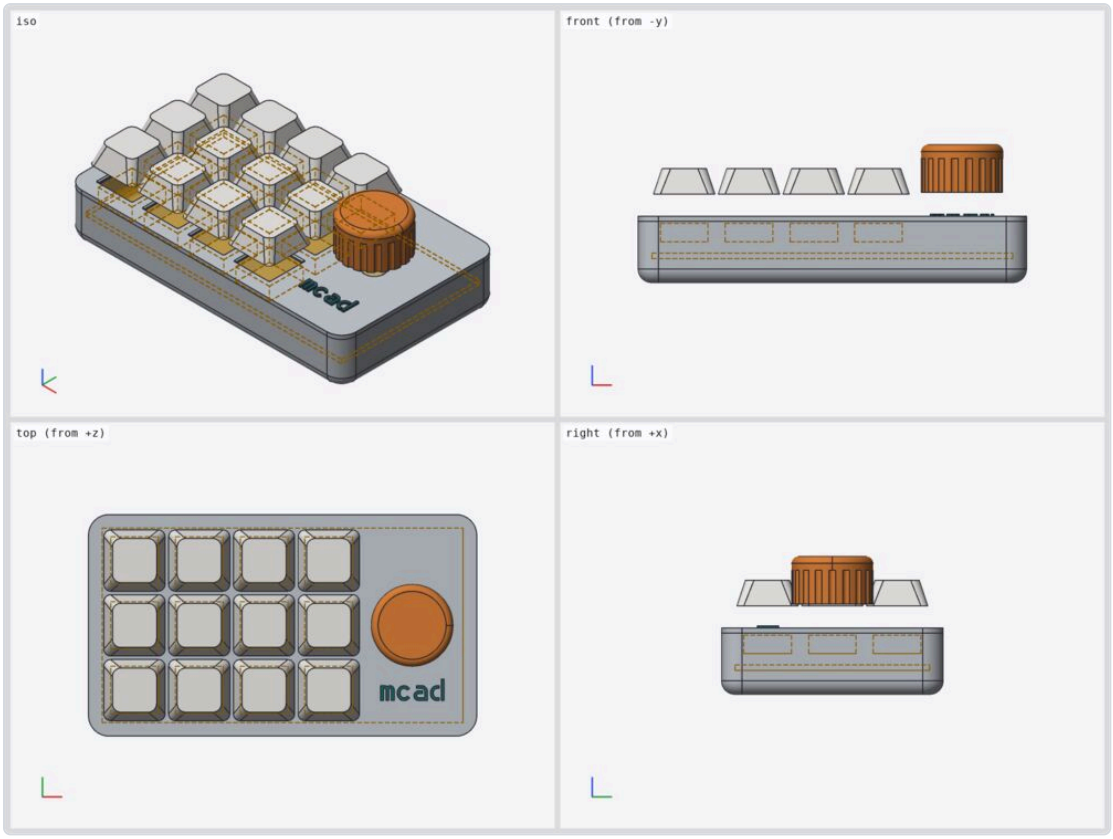


`1`: wide penumbra far from the contact, sharp at it

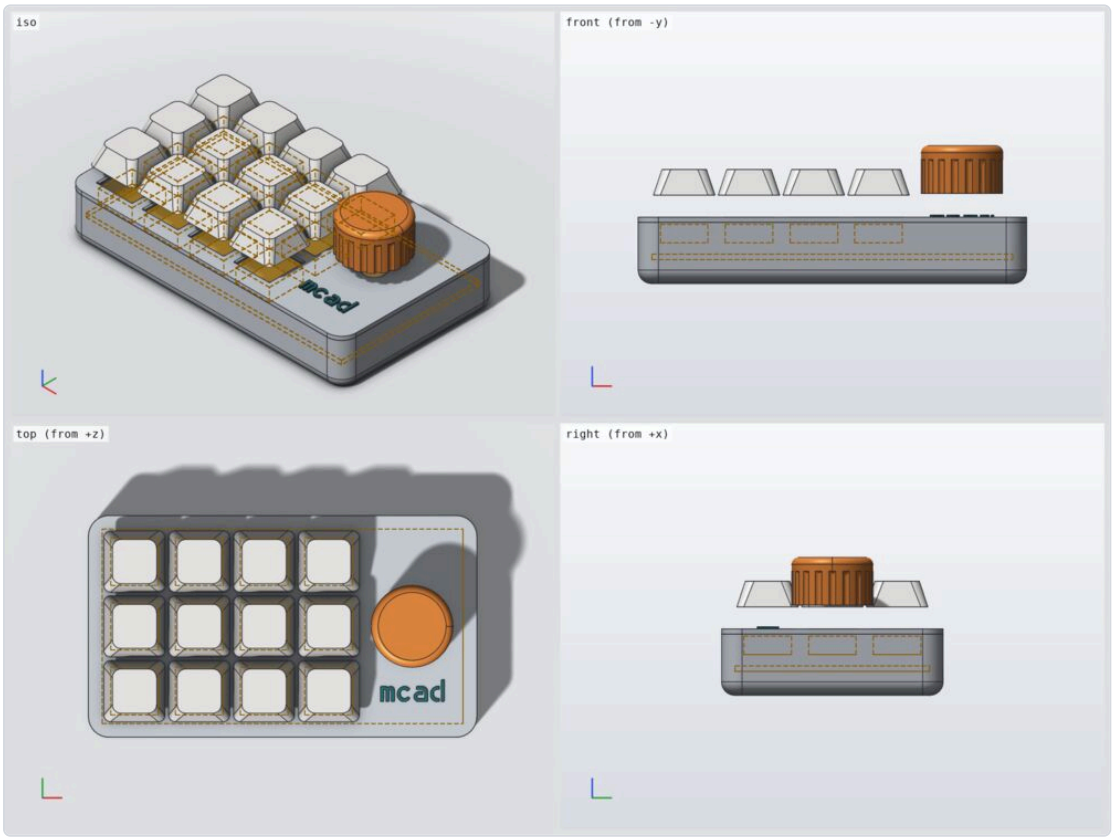


`--light-intensity 1.4 --ambient 0.2`: more contrast

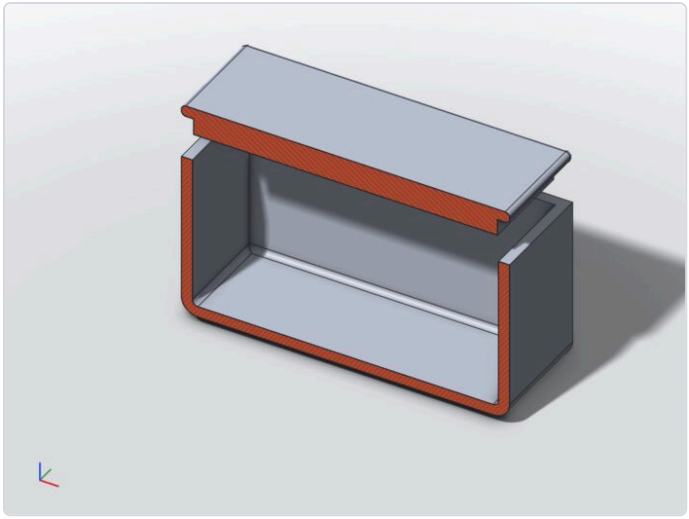
Contact sheet, section, perspective, x-ray



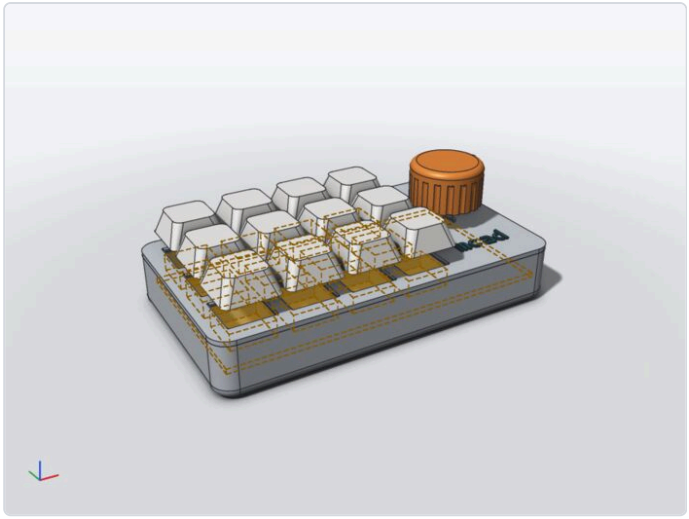
`--views all`, standard



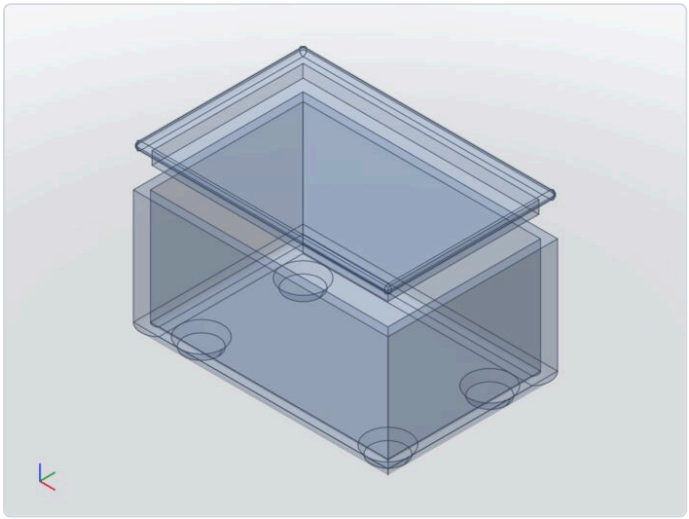
`--views all --style studio`: same scale per tile; the light follows each tile's camera, the ground only appears where the camera looks down (iso, top)



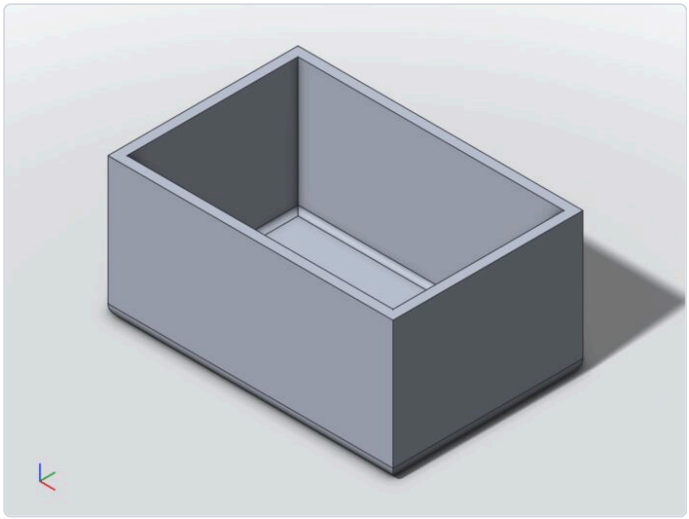
`--section y=20 --section-flip`: caps lit and shadowed, interior with AO



`--perspective --azimuth -30 --elevation 28`



`--xray`: transparent bodies cast no shadow, the ground stays



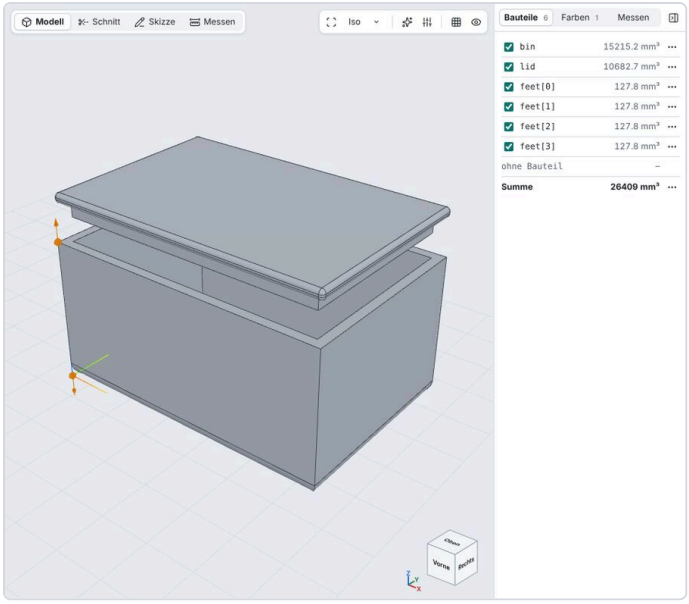
`--part bin`

4 UI screenshots

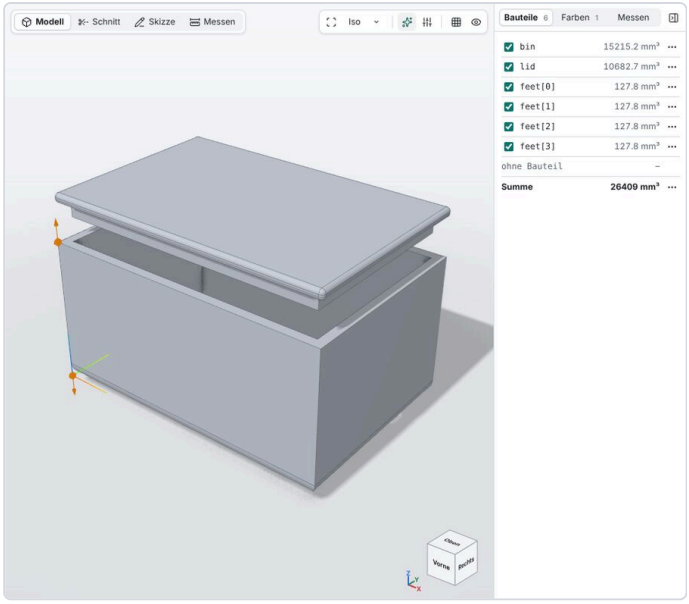
Browser build with the Wasm kernel in Chromium (software WebGL via SwiftShader), window 1440×900; German UI as in all Playwright specs (“Studio-Darstellung” = studio look, “Studio-Einstellungen” = studio settings, “Umgebungsverdeckung” = ambient occlusion, “Schatten” = shadows, “Boden” = ground, “Hintergrund” = background, “Standardwerte” = defaults). The desktop app uses the same web UI.

S01 Studio look on and off

The button “Studio-Darstellung” (studio look) in the view toolbar switches to key/fill/sky light, soft shadows on the model and the ground, ambient occlusion and a fading ground on a gradient background; 50 % of the view pixels change. Frame time (software WebGL): standard 55.8 ms, studio 1932.7 ms, studio without AO 1395.3 ms.



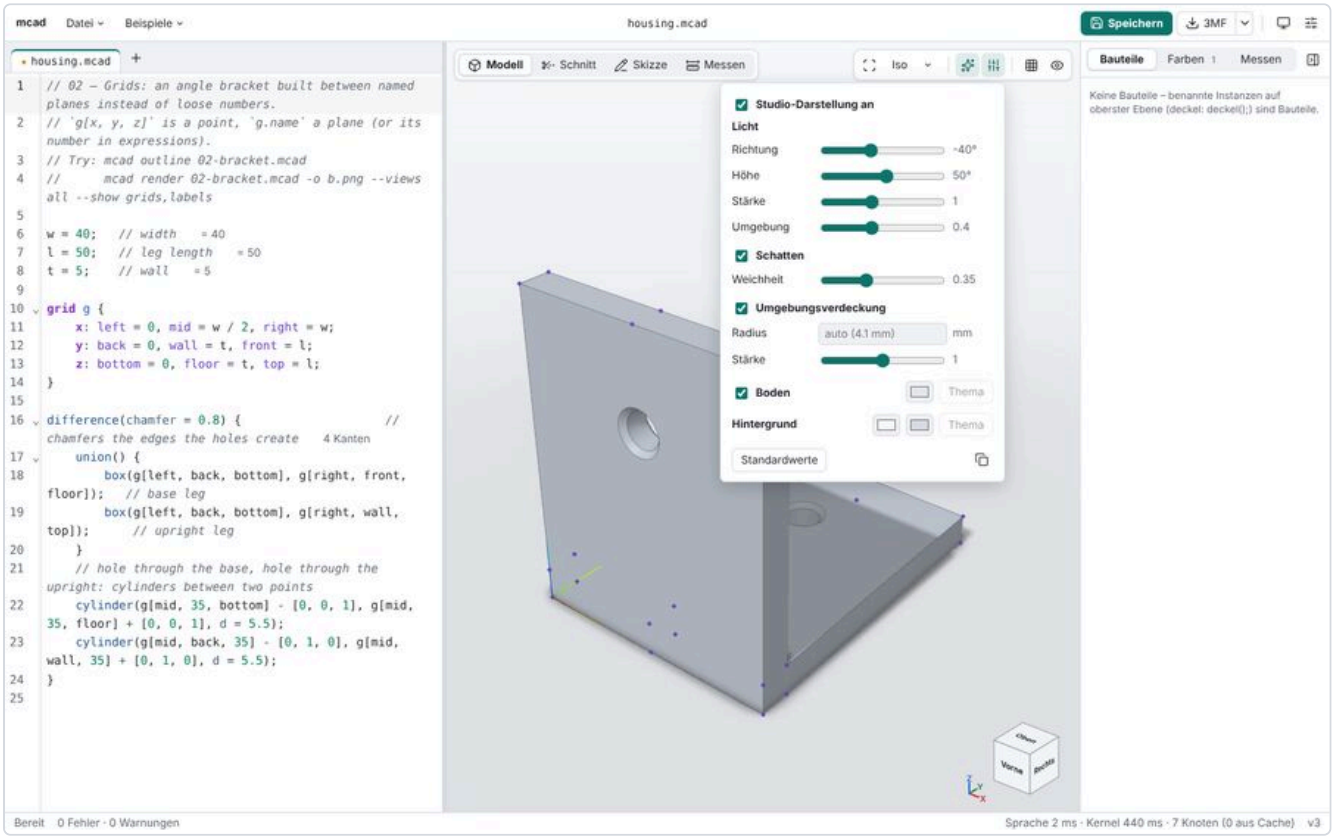
S01: standard



S01: studio look

S02 Settings panel and persistence

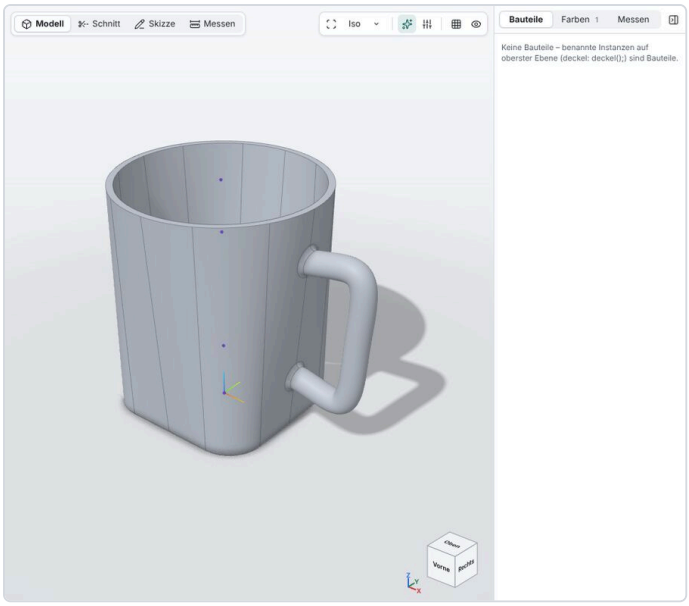
Panel “Studio-Einstellungen”: switching off ambient occlusion changes 1.2 % of the pixels, switching off shadows another 3.4 %; light direction 60°/25° is used (light vector z = sin 25°); after a reload the settings are still there (localStorage `cad-studio`); “Standardwerte” restores the defaults.



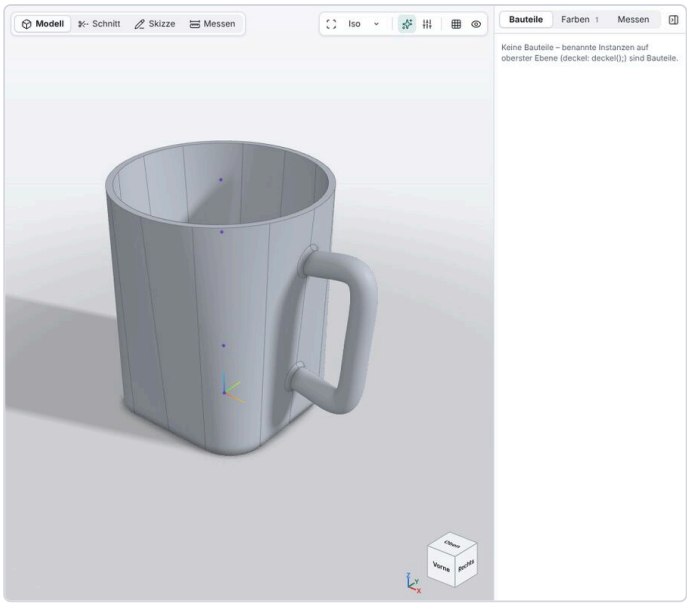
S02: settings panel

S03 Variants

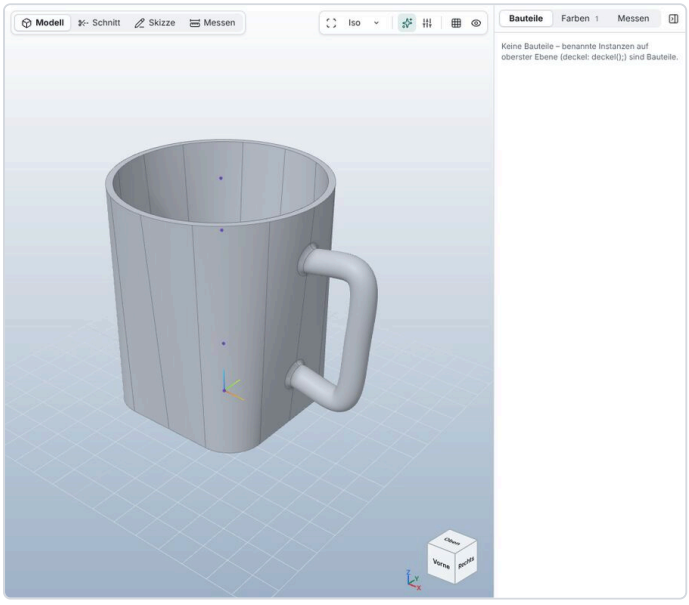
Mug: default settings; light from the right at 30° with soft shadows (0.9); ground off on a white–blue gradient; dark theme (ground and gradient from the theme).



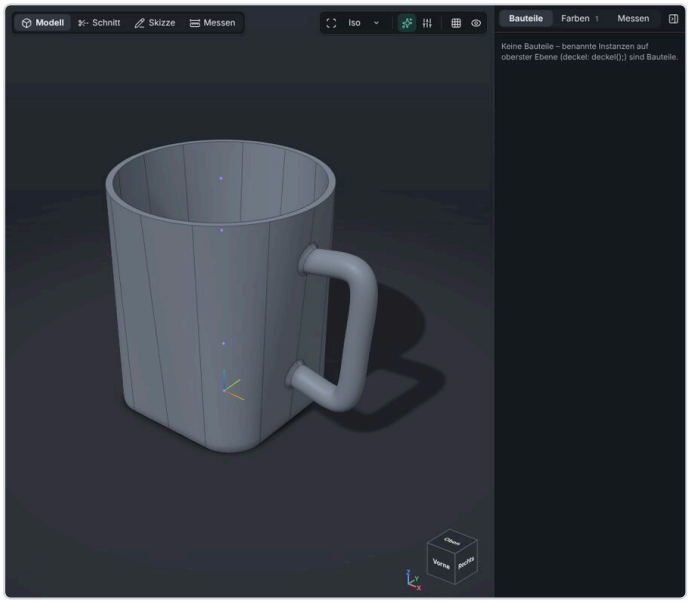
S03: default



S03: light from the right at 30°, softness 0.9



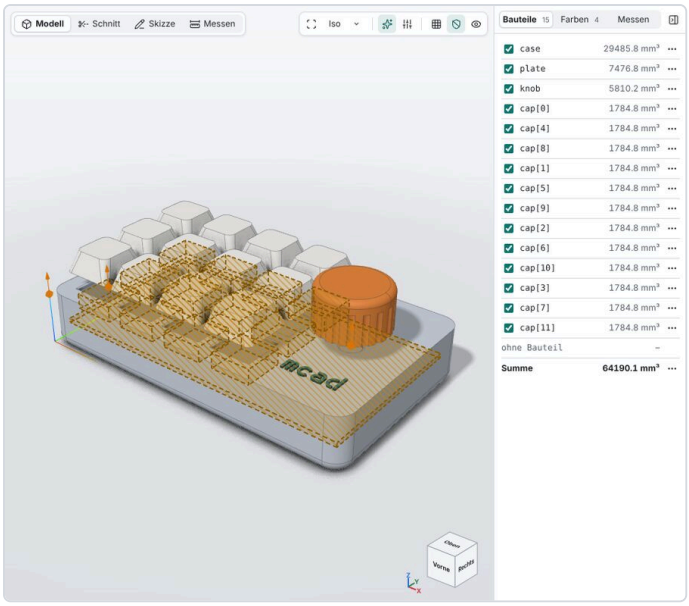
S03: ground off, white–blue gradient



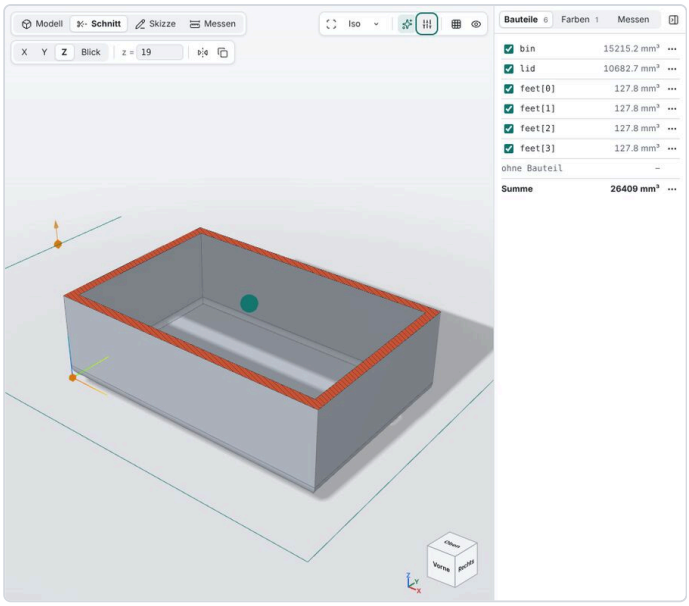
S03: dark theme

S04 Section and further models

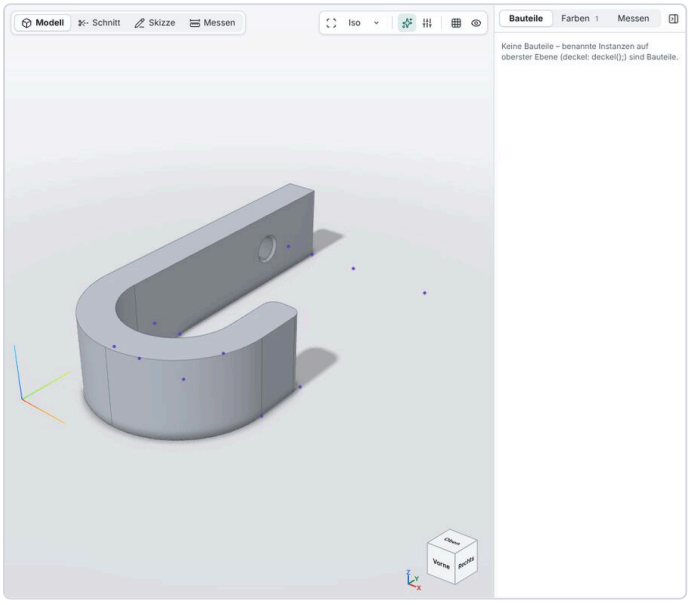
Macropad (colours per part, keep-out volumes) in the studio look; the box with a section (caps, shadows; ambient occlusion pauses during a section, the panel says so); hook.



S04: macropad



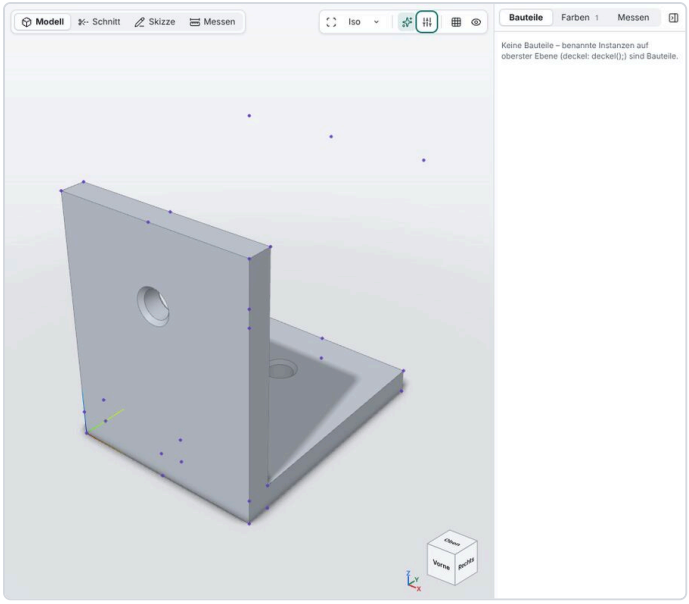
S04: section (AO paused)



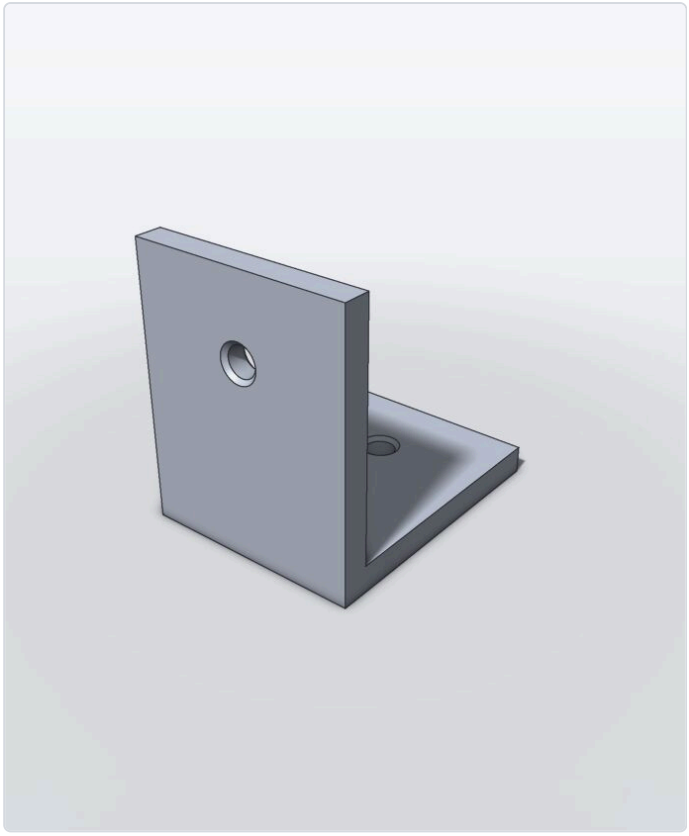
S04: hook

S05 Same look in the CLI

“Als mcad render kopieren” gives `mcad render housing.mcad -o studio.png --perspective --azimuth 38.7 --elevation 29.3 --style studio --light -30,50 --shadow-softness 0.5`; the call run with the CLI (size of the view, --no-axes) renders the same view in the CLI's software renderer.



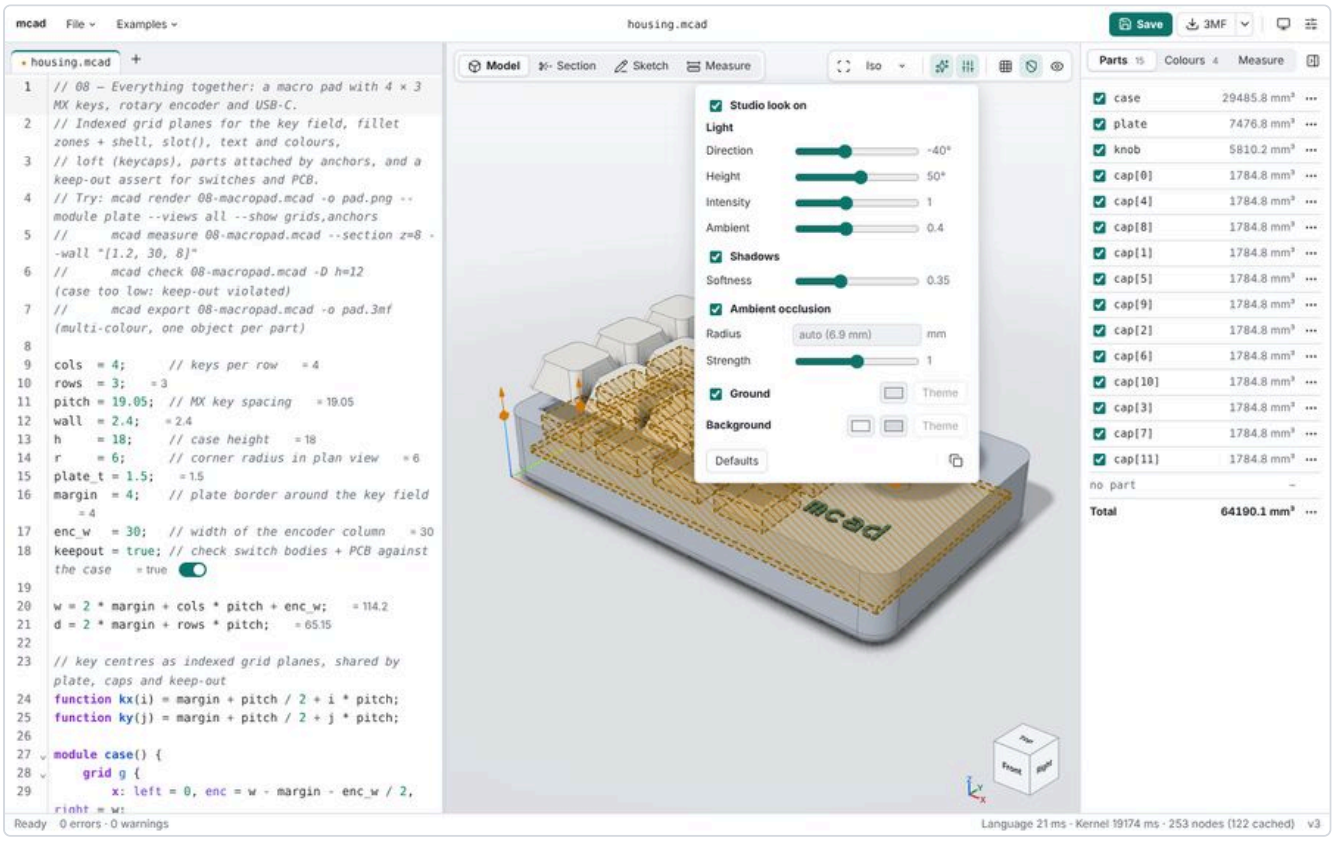
S05: UI



S05: CLI with the copied call

S06 English UI

Browser language English: “Studio look” and “Studio settings” in the view toolbar, the panel with Light (Direction, Height, Intensity, Ambient), Shadows (Softness), Ambient occlusion (Radius, Strength), Ground (colour, Theme), Background (top, bottom, Theme), Defaults and “Copy as mcad render”.



S06: settings panel, English UI

5 Tests

New tests

Where	What
`crates/cad-render/tests/studio.rs` (7)	ground drawn and studio differs from standard, deterministic, ground off → corner is background; ground behind a wall in its shadow (≥ 40 brightness levels darker), lit side and far ground unchanged, wider transition with softness 1; plate shadows the block below; AO darkens a concave corner (≥ 8 levels) but not the open floor, strength 2 darker than 0.5; ground only from above, light azimuth relative to the camera (front/right/top), all views in perspective; gradient background; transparent bodies cast no shadow
`crates/cad-cli/tests/agent.rs` `render_studio_style`	JSON defaults (`style`, `studio`, `background`, `views[].studio`), every option parsed and reported, images differ, contact sheet + dark + section + perspective together, gradient background with standard (top white, bottom black), text line, 10 usage errors (exit 2)
`web/src/logic/studio.test.ts` (7)	defaults equal to the CLI, sanitising stored settings, AO radius, light direction (same cases as the Rust test), CLI arguments and full command
`web/e2e/studio.spec.ts` (S01–S06)	toggle, panel, persistence after reload, defaults, variants, dark theme, section, copied CLI call run with the CLI, English labels

Rust studio tests in the final run: `render_studio_style` ok, `gradient_background` ok, `model_shadows_itself` ok, `shadow_darkens_the_ground_behind_a_wall` ok, `ambient_occlusion_darkens_a_concave_corner` ok, `transparent_items_cast_no_shadow` ok, `studio_draws_ground_and_differs_from_standard` ok, `ground_only_from_above_and_lights_follow_the_camera` ok.

Test runs on the final state

Command	Result	Duration
`cargo fmt --all --check`	**passed** – no diff	1 s
`cargo clippy --workspace --exclude cad-app --no-default-features --all-targets -- -D warnings`	**passed** – no findings	23 s
`cargo test --workspace --exclude cad-app --no-default-features --no-fail-fast`	**passed** – 695 passed, 0 failed, 0 ignored (109 test binaries)	6.0 min
`cargo test -p cad-cli usage` (manual: titles, every command)	**passed** – 7 passed, 0 failed	8 s
`cd web && pnpm typecheck`	**passed** – no errors	1 s
`cd web && pnpm test` (Vitest)	**passed** – 271 passed, 0 failed	4 s
Playwright 1.56.1, full suite (Chromium 1243 with SwiftShader, `playwright.local.config.ts`)	–	–

The container has no WebKitGTK: the desktop app crate `cad-app` and the `gui` feature of `cad-cli` are not built (the UI is the same web build, tested in Chromium). Playwright 1.56.1 expects Chromium 1194; the preinstalled Chromium 1243 is used through the gitignored `web/playwright.local.config.ts` (only `executablePath` differs). Logs in `reports/tmp/report22/final/`.

6 Performance

CLI: `render_ms` of `--json`, i.e. everything after the evaluation (tessellation, colour regions of coloured models, rasterising with 2×2 supersampling, studio pass, PNG encoding; evaluation and kernel excluded), 1024×768, median of 5 runs, 4 cores. The macropad is dominated by its colour regions (Booleans per colour, about 6 s in both styles).

Model	before (standard, ms)	standard	studio	studio, no shadows	studio, no AO	studio, neither	studio / standard
`02-bracket`	202	193	**1,138**	771	789	363	5.9×
`03-tray`	337	359	**1,301**	919	893	525	3.6×

Model	before (standard, ms)	standard	studio	studio, no shadows	studio, no AO	studio, neither	studio / standard
`04-hook`	167	185	**1,142**	773	581	339	6.2×
`05-box-lid`	421	492	**1,390**	1,000	1,072	649	2.8×
`07-mug`	953	879	**2,016**	1,474	1,443	1,039	2.3×
`08-macropad`	5,854	6,408	**7,400**	6,593	6,733	5,952	1.2×
macropad `--views all` 1600×1200		6,338	**7,665**				1.2×
box-lid 1920×1440		1,468	**4,509**				3.1×

The studio pass adds a roughly constant 0.9–1.1 s at 1024×768 (shadow map and lookups ≈ 0.4 s, AO ≈ 0.4 s, deferred lighting ≈ 0.15 s), so it weighs most on light models and scales with the pixel count (1920×1440: see the last row); the standard style is as fast as before the change (column “before”: the build of the commit before the change).

UI (Playwright S01, Chromium, SwiftShader (software WebGL), 1440×900 window, mean of 8 synchronous frames incl. `readPixels`): standard 55.8 ms, studio 1,932.7 ms, studio without AO 1,395.3 ms per frame. Software WebGL is far slower than a GPU; the numbers only compare the modes.

7 Known limitations

Item	Cause or status
Screen-space AO	Only what is on screen occludes (in CLI and UI); very large radii darken silhouettes against far geometry less correctly. The UI's GTAO shows slight noise on software WebGL; the CLI is noise-free after supersampling and blur.
AO in the UI with a section	paused while a section is shown: the cut faces are stencil caps that the AO normal pass cannot reproduce (the panel says so). The CLI computes AO with sections.
Soft shadows differ in detail	CLI: percentage-closer soft shadows (penumbra grows with the distance to the occluder). UI: variance shadow map with a constant blur radius derived from the same softness; contact hardening only in the CLI.
Shadow casters	transparent bodies (x-ray, dimmed rest, alpha) and section caps cast no shadow; the shadow map covers the model, very long shadows at low light elevation end where the ground fades.
Ground	only below the model and only when the camera looks down onto it; not in front/right/bottom views. A model “floating” above other parts is lit as one scene (no per-part ground).
Light follows the camera azimuth	deliberately (turntable): every view is lit from the front-left; in the UI the shadow turns while orbiting.
Run time	studio is 3.6× slower than standard in the CLI (table above); `--shadows off --ao off` keeps only the lighting.
Desktop app	not built or run here (no WebKitGTK); it embeds the same web UI.